Math 4261: Mathematical Statistics I

Lecture 19: Joint distributions of estimators, chi-squared and t-distributions
D-S: 8.1–8.4

Fall 2026

Outline

Joint distribution of estimators in Gaussian model

Student-t distribution

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Joint distribution of estimators in Gaussian model

Student-t distribution

Setting and motivation

As motivated by the misspecified MLE asymptotic variance: we often find ourselves having to estimate the mean and variance of particular statistics.

Given data $X_1, \ldots, X_n$ from some distribution, a standard estimate for the true mean of the distribution is the sample mean

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

We have worked with this quantity quite substantially. Results like the LLN and CLT provide an understanding of its distribution.

A standard estimate for the variance is the sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

A reason for the $\frac{1}{n-1}$ is that $\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2\right] = \sigma^2\left(\frac{n-1}{n}\right)$.

Sample mean and Sample variance

We will study the joint distribution of $\overline{X}$ and S^2 in the **Gaussian** case, i.e. when $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Here are several motivations:

- The Gaussian is a common and often well-justified modeling assumption.
- We sometimes assume σ^2 is known (for instance to evaluate $I(\mu) = \frac{1}{\sigma^2}$). However, in many situations, we will have to estimate the mean and variance jointly.
- We can make precise statements in this case.
- (An excuse for me to introduce the χ^2 and t-distributions).

Joint distribution of estimators

Theorem

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, where $n \geq 2$ and $\sigma^2 > 0$. Then

(i) $\frac{\sqrt{n}}{\sigma}\left(\overline{X} - \mu\right) \sim \mathcal{N}(0,1)$.

(ii) S^2 is an unbiased estimator for σ^2

(iii) $S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$

(iv) S^2 and $\overline{X}$ are independent.

Statement (i) follows because linear functions of Gaussians are Gaussians. We have done statement (ii) in Lecture 8.

We will prove (iii) and (iv) now.

Step 0: Reduction

Theorem

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, where $n \geq 2$ and $\sigma^2 > 0$. Then

(iii) $S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$

(iv) S^2 and $\overline{X}$ are independent.

Reduction: It suffices to prove this for $\mu = 0$ and $\sigma^2 = 1$.

Let $Z_1, \ldots Z_n \overset{iid}{\sim} N(0,1)$. Let $\bar{Z}$: sample mean, S_Z^2 : sample variance

Suppose $\bar{Z}$ independent of S_Z^2 and $S_Z^2 \sim \frac{1}{n-1}\chi^2_{n-1}$.

Represent $X_i = \mu + \sigma Z_i \Rightarrow \bar{X} = \mu + \sigma \bar{Z}$, $S^2 = \sigma^2 S_Z^2$.

Then $\bar{X}$ independent of S^2, and $S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$.

Independence of sample mean and variance

Assume $\mu = 0$, $\sigma^2 = 1$,
so $X_1, \dots, X_n \overset{iid}{\sim} N(0,1)$

Proof #1: (iv) S^2 and $\overline{X}$ are independent.

:) Consider $Y = (\overline{X}, X_1 - \overline{X}, X_2 - \overline{X}, \dots, X_n - \overline{X})$.

This is a linear transformation of $X = (X_1 \dots X_n)$, so it is a multivariate normal.

The specific linear transformation A:

$$Y^{(n+1)\times 1} = A^{(n+1)\times n} X^{n\times 1}$$

$$\begin{bmatrix} \overline{X} \\ X_1 - \overline{X} \\ X_2 - \overline{X} \\ \vdots \\ X_n - \overline{X} \end{bmatrix} = \begin{bmatrix} \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} \\ 1-\frac{1}{n} & -\frac{1}{n} & \cdots & -\frac{1}{n} \\ -\frac{1}{n} & 1-\frac{1}{n} & \cdots & -\frac{1}{n} \\ \vdots & & & \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ \vdots \\ X_n \end{bmatrix}$$

$$(1-\tfrac{1}{n})X_1 - \tfrac{1}{n}X_2 - \cdots - \tfrac{1}{n}X_n = X_1 - \tfrac{1}{n}(X_1 + \cdots + X_n) = X_1 - \overline{X}$$

Independence of sample mean and variance

Proof #1: (iv) S^2 and $\overline{X}$ are independent.

Assume $\mu=0$, $\sigma^2=1$, so $X_1,\dots,X_n \overset{iid}{\sim} N(0,1)$

·) Consider $(\bar{X}, X_1-\bar{X}, X_2-\bar{X}, \dots, X_n-\bar{X})$.

This is a linear transformation of $(X_1 \dots X_n)$, so it is a multivariate normal.

·) $\mathrm{Cov}[\bar{X}, X_i-\bar{X}] = \underbrace{\mathrm{Cov}[\bar{X}, X_i]}_{\text{I}} - \underbrace{\mathrm{Cov}[\bar{X}, \bar{X}]}_{\text{II}}$

$\text{I} = \mathrm{Cov}\left[\frac{1}{n}(X_1+\dots+X_n), X_i\right]$

$= \frac{1}{n}\sum_{j=1}^{n} \underbrace{\mathrm{Cov}[X_j, X_i]}_{=0 \text{ unless } j=i} = \frac{1}{n}\mathrm{Var}[X_i] = \frac{1}{n}$

Independence of sample mean and variance

Proof #1: (iv) S^2 and $\overline{X}$ are independent.

Assume $\mu = 0$, $\sigma^2 = 1$,
so $X_1, \ldots, X_n \overset{iid}{\sim} N(0,1)$

$$\text{II} = \text{Cov}[\bar{X}, \bar{X}] = \text{Var}\,\bar{X} = \frac{1}{n}$$

Hence $\text{Cov}[\bar{X}, X_i - \bar{X}] = 0$ and so $\bar{X}$ is uncorrelated with each of $X_1 - \bar{X}, \ldots, X_n - \bar{X}$.

Further conclude, by multivariate normal properties, that $\bar{X}$ independent of $(X_1 - \bar{X}, \ldots, X_n - \bar{X})$.

So $\bar{X}$ independent of S^2.

Recap: Rotational symmetry of std MVN

Let $\boldsymbol{Z} = (Z_1, \ldots, Z_p) \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_{p\times p})$ and let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^p$ be any two vectors that are perpendicular to each other:

$$\boldsymbol{a}^\top \boldsymbol{b} = 0$$

$$\begin{bmatrix} a^\top Z \\ b^\top Z \end{bmatrix} = \begin{bmatrix} - a^\top - \\ - b^\top - \end{bmatrix} \begin{bmatrix} | \\ Z \\ | \end{bmatrix}$$

Then $\boldsymbol{a}^\top \boldsymbol{Z}$ and $\boldsymbol{b}^\top \boldsymbol{Z}$ are **independent**, with

$$\boldsymbol{a}^\top \boldsymbol{Z} \sim \mathcal{N}(0, \|\boldsymbol{a}\|^2), \qquad \boldsymbol{b}^\top \boldsymbol{Z} \sim \mathcal{N}(0, \|\boldsymbol{b}\|^2).$$

$$\sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \star & \cdot \\ \cdot & \star \end{bmatrix} \right)$$

In particular, when $\boldsymbol{a}$ and $\boldsymbol{b}$ are unit vectors, then $\boldsymbol{a}^\top \boldsymbol{Z}$ and $\boldsymbol{b}^\top \boldsymbol{Z}$ are i.i.d $\mathcal{N}(0, 1)$.

$$\star = \mathrm{Cov}[a^\top Z, b^\top Z] = \mathbb{E}[a^\top Z b^\top Z] - \mathbb{E}[a^\top Z]\,\mathbb{E}[b^\top Z]$$

$$= \mathbb{E}[a^\top Z Z^\top b] = a^\top \underbrace{\mathbb{E}[ZZ^\top]}_{I_{p\times p}} b = a^\top b = 0$$

Independence of sample mean and variance

Proof #2: (iv) S^2 and $\overline{X}$ are independent.

Assume $\mu = 0$, $\sigma^2 = 1$, so $X_1, \ldots, X_n \overset{iid}{\sim} N(0,1)$

Let $X = (X_1, \ldots, X_n) \sim N(0, I_{n \times n})$ std. MVN

$$\overline{X} = \underbrace{\left(\frac{1}{n}, \ldots, \frac{1}{n}\right)^T}_{u} X, \quad X_1 - \overline{X} = \underbrace{\left(1 - \frac{1}{n}, -\frac{1}{n}, \ldots, -\frac{1}{n}\right)^T}_{v_1} X$$

$$u^T v_1 = \frac{1}{n}\left(1 - \frac{1}{n}\right) \underbrace{- \frac{1}{n^2} - \cdots - \frac{1}{n^2}}_{n-1 \text{ times}} = \frac{1}{n} - n \cdot \frac{1}{n^2} = 0$$

So u perpendicular to v_1. Length of $u = \|u\| = \frac{1}{\sqrt{n}}$

Define $X_i - \overline{X} = v_i^T X$. Similarly u perpendicular to v_i.

Independence of sample mean and variance

Proof #2: (iv) S^2 and $\overline{X}$ are independent.

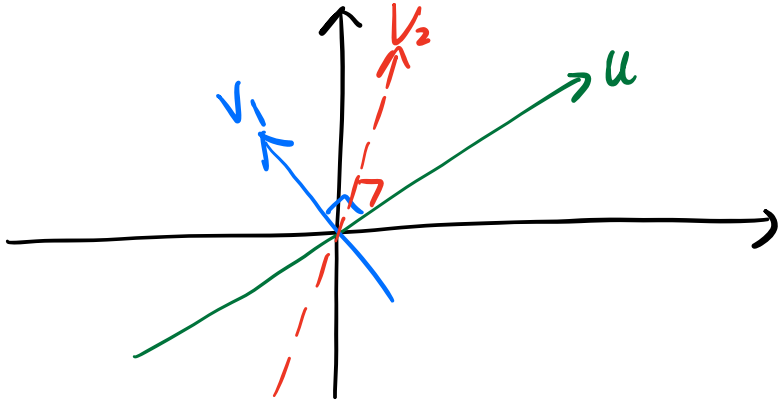
Assume $\mu=0$, $\sigma^2=1$,
so $X_1, \ldots, X_n \overset{iid}{\sim} N(0,1)$

By rotational symmetry of MVN,

$\overline{X}$ and $(X_1-\overline{X}, \ldots, X_n-\overline{X})$ have same distribution as $\frac{1}{n}X_1$ and some combination of $X_2, \ldots, X_n$

$\frac{1}{n}X_1$ → these are independent

$\Rightarrow \overline{X}$ and S^2 independent



Recap: Chi-squared distribution

A chi-squared random variable $X \sim \chi^2_n$ is a special case of a Gamma random variable, Gamma$(n/2, 1/2)$. PDF:

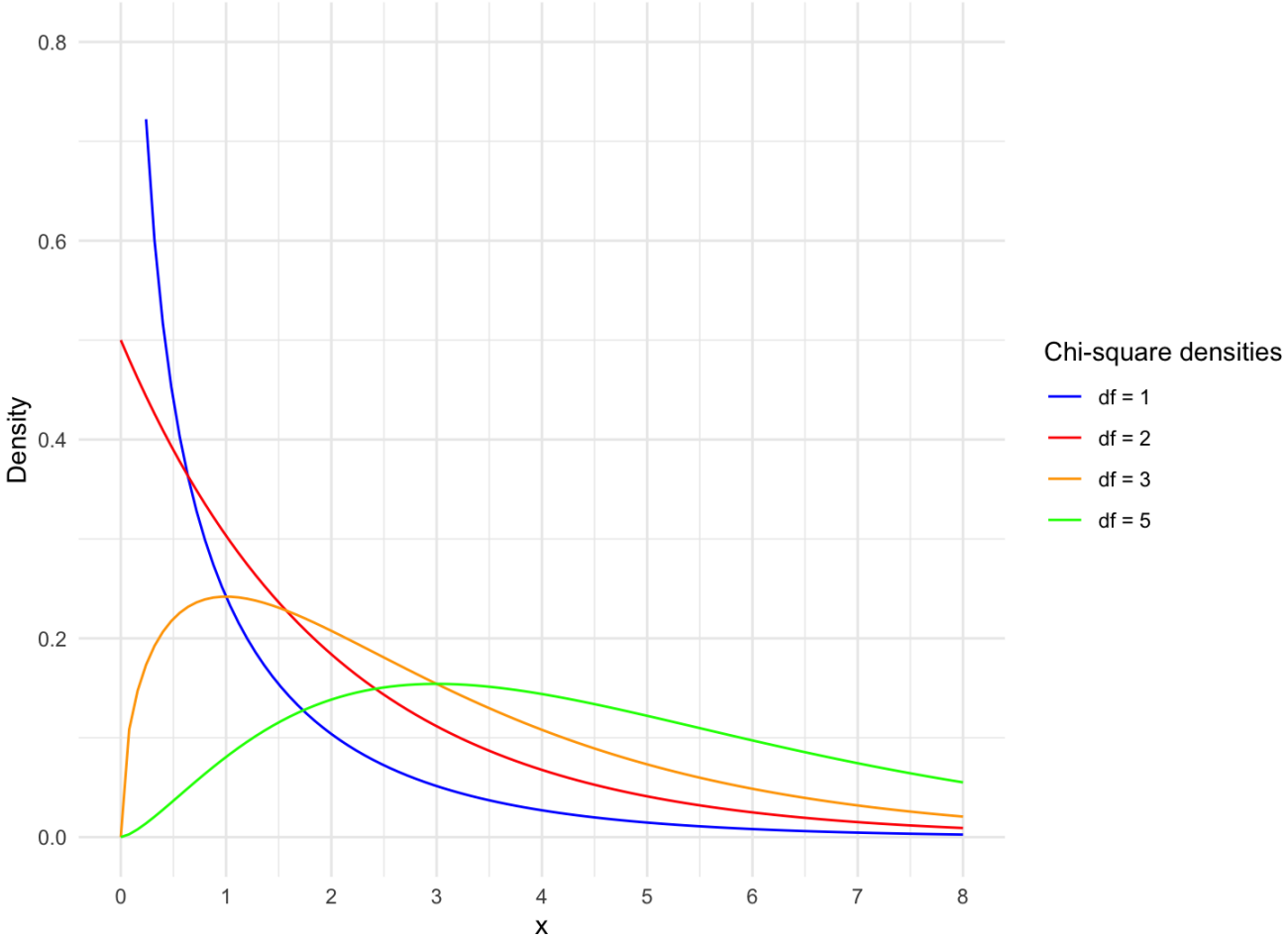
$$f_X(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}, \qquad \text{for } x > 0.$$

The parameter n is called the "degrees of freedom".

Useful representation (often more so than PDF):
If $Z_1, \ldots, Z_n \sim \mathcal{N}(0,1)$ i.i.d., then

$$Z_1^2 + Z_2^2 + \cdots + Z_n^2 \sim \chi^2_n$$

Chi-squared distribution



Distribution of sample variance

Assume $\mu = 0$, $\sigma^2 = 1$, so $X_1, \dots, X_n \overset{iid}{\sim} N(0,1)$

Proof: (iii) $S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$

- Note $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 = \frac{1}{n-1}\left[\underbrace{X_1^2 + \dots + X_n^2}_{W} - \underbrace{n\bar{X}^2}_{V}\right]$

- Let $U = (n-1)S^2$, $W = X_1^2 + \dots + X_n^2$, $V = n\bar{X}^2$

  so $U = W - V \Rightarrow W = U + V$

- By (iii), S^2 and $\bar{X}$ independent, so U and V independent

- $W \sim \chi^2_n$, $\sqrt{n}\,\bar{X} \sim N(0,1) \Rightarrow V = (\sqrt{n}\,\bar{X})^2 \sim \chi^2_1$

  "$\chi^2_n = U + \chi^2_1$" $\Rightarrow$ This implies $U \sim \chi^2_{n-1}$

  Hence $S^2 = \frac{1}{n-1}U \sim \frac{1}{n-1}\chi^2_{n-1}$.

Example

(D-S 8.3 page 475). Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, where both parameters are unknown and to be estimated. Suppose we use the MLE's $\widehat{\mu} = \overline{X}$ and $\widehat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2}$ to estimate μ and σ. Find the smallest sample size n so that the following relation is satisfied:

$$(*) \quad \mathbb{P}\left[|\widehat{\mu} - \mu| \le \frac{1}{5}\sigma \quad \text{and} \quad |\widehat{\sigma} - \sigma| \le \frac{1}{5}\sigma \right] \ge \frac{1}{2}$$

Note $\widehat{\sigma}^2 = \frac{1}{n}\sum_i (X_i - \overline{X})^2 = \frac{n-1}{n} S^2$

S^2 and $\overline{X}$ indpt $\Rightarrow$ $\widehat{\sigma}$ and $\widehat{\mu}$ indpt.

$$(*) = \underbrace{\mathbb{P}\left[|\widehat{\mu} - \mu| \le \frac{1}{5}\sigma \right]}_{p_1} \cdot \underbrace{\mathbb{P}\left[|\widehat{\sigma} - \sigma| \le \frac{1}{5}\sigma \right]}_{p_2}$$

Example

$$p_1 = P\left[|\bar{X}-\mu| \le \tfrac{1}{5}\sigma\right] = P\left[\tfrac{\sqrt{n}}{\sigma}|\bar{X}-\mu| \le \tfrac{\sqrt{n}}{5}\right]$$

$$= P\left[N(0,1) \le \tfrac{\sqrt{n}}{5}\right]$$

$$p_2 = P\left[|\hat{\sigma}-\sigma| \le \tfrac{1}{5}\sigma\right] = P\left[\tfrac{4}{5} \le \tfrac{\hat{\sigma}}{\sigma} \le \tfrac{6}{5}\right]$$

$$= P\left[0.64 \le \frac{n-1}{n}\frac{S^2}{\sigma^2} \le 1.44\right] \qquad \frac{n-1}{n}\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$$

$$= P\left[0.64n \le \chi^2_{n-1} \le 1.44n\right]$$

Numerically evaluate for $n = 1, 2, \ldots$

For $\boxed{n=21}$, $p_1 = 0.64$, $p_2 = 0.78$ $\Rightarrow$ $p_1 p_2 = 0.5$

Outline

Joint distribution of estimators in Gaussian model

Student-t distribution

Student-t distribution

The ‘Student’ is William Sealy Gosset.





VOLUME VI MARCH, 1908 No. 1

BIOMETRIKA.

THE PROBABLE ERROR OF A MEAN.

BY STUDENT.

Introduction.

ANY experiment may be regarded as forming an individual of a "population" of experiments which might be performed under the same conditions. A series of experiments is a sample drawn from this population.

Now any series of experiments is only of value in so far as it enables us to form a judgment as to the statistical constants of the population to which the experiments belong. In a great number of cases the question finally turns on the value of a mean, either directly, or as the mean difference between the two quantities.

If the number of experiments be very large, we may have precise information

Source: Wikipedia and `https://www.ecoevoblog.com/2020/10/20/guinness-and-t-tests-to-gosset/`

The Student-t distribution

Definition

Let Z and U be independent random variables with $Z \sim \mathcal{N}(0,1)$ and $U \sim \frac{1}{n}\chi^2_n$. Then the distribution of

$$T = \frac{Z}{\sqrt{U}}$$

is called the **Student-t distribution** with n degrees of freedom.

We write

$$T \sim t_n.$$

Example: For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, the so-called t-statistic

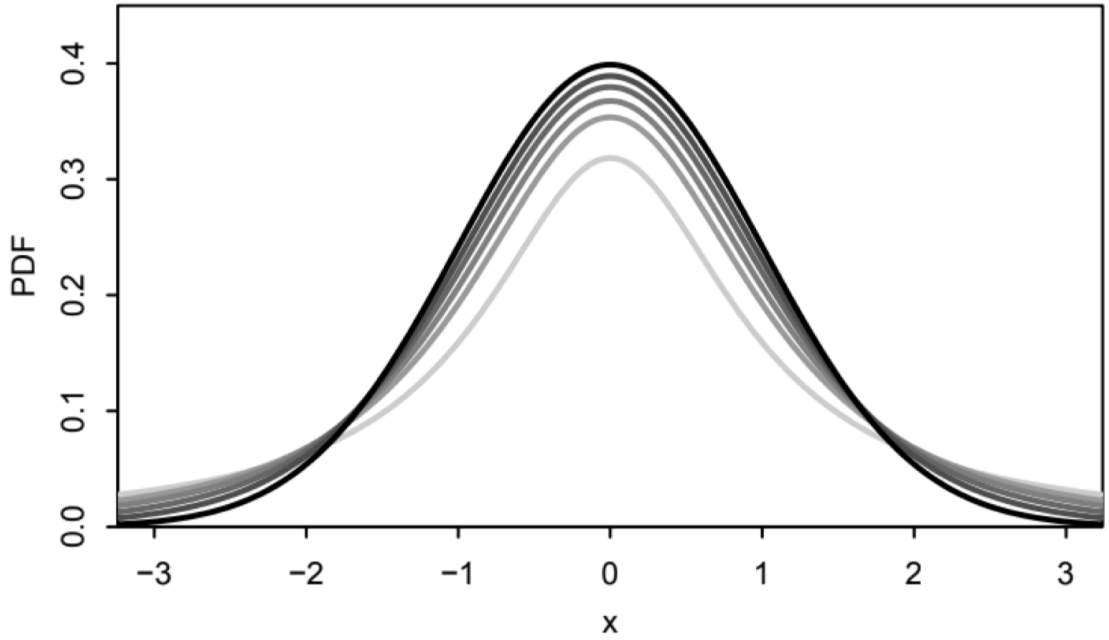
$$T = \frac{\sqrt{n}}{S}\left(\overline{X} - \mu\right) = \frac{\frac{\sqrt{n}}{\sigma}\left(\overline{X} - \mu\right)}{\sqrt{\frac{S^2}{\sigma^2}}} \sim t_{n-1}$$

$\frac{\sqrt{n}}{\sigma}(\overline{X} - \mu) \sim \mathcal{N}(0,1)$

$\frac{S^2}{\sigma^2} \sim \frac{1}{n-1}\chi^2_{n-1}$

$S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$

The Student-t distribution



PDF of the t-distribution t_n with degrees-of-freedom $n = 1, 2, 3, 5, 10$ (light to dark).

t-distribution converges to Gaussian

Claim

Let $Z \sim \mathcal{N}(0,1)$ and $U \sim \frac{1}{n}\chi^2_n$ be independent random variables so that $T = Z/\sqrt{U} \sim t_n$. Then as $n \to \infty$,

$$T \xrightarrow{d} \mathcal{N}(0,1)$$

proof: " $T = \frac{\mathcal{N}(0,1)}{\sqrt{\frac{1}{n}\chi^2_n}}$ " . Suffices to show $U \xrightarrow{\text{in prob}} 1$

$U = \frac{1}{n}\sum_{i=1}^{n} W_i^2$ for $W_i \overset{iid}{\sim} \mathcal{N}(0,1)$

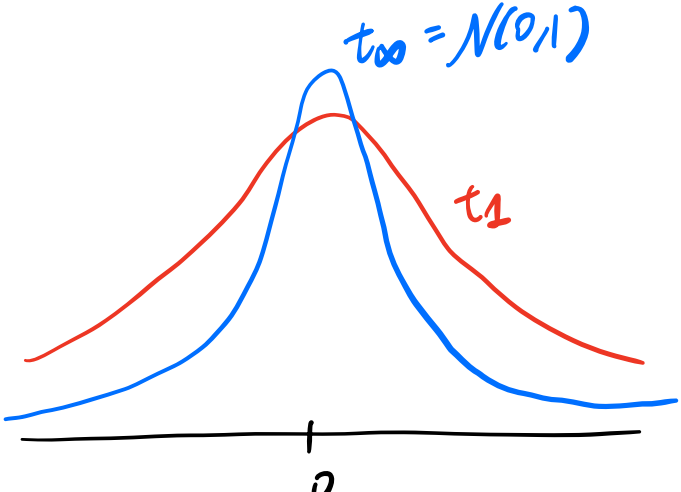
$\xrightarrow{LLN} \mathbb{E}[W_i^2] = \operatorname{Var} W_i = 1$.

The (evil) Cauchy distribution

The t distribution with one degree of freedom, t_1, is the distribution of $Y = Z/\sqrt{U}$ where $Z \sim \mathcal{N}(0,1)$ and $U \sim \chi^2_1$ are independent.

Fact: t_1 coincides with the standard Cauchy distribution which has PDF

$$f_Y(y) = \frac{1}{\pi(1+y^2)}, \quad y \in \mathbb{R}$$



$$\mathbb{E}Y \simeq \frac{1}{\pi}\int_{\mathbb{R}} y \cdot \frac{1}{y^2}\, dy$$

blows up.

The mean and variance of this distribution do not exist.

Example

(D-S Exercise 8.4.2). Suppose that $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ where μ and σ are unknown. Let $\widehat{\mu}$ and $\widehat{\sigma}$ denote the MLE's of μ and σ. For the sample size $n = 17$, find a value of k such that

$$\mathbb{P}\left[\widehat{\mu} > \mu + k\widehat{\sigma}\right] = 0.95.$$

Note $\hat{\mu} = \bar{X}$, $\hat{\sigma}^2 = \frac{n-1}{n} S^2 \Rightarrow \hat{\sigma} = \sqrt{\frac{n-1}{n}}\, S$

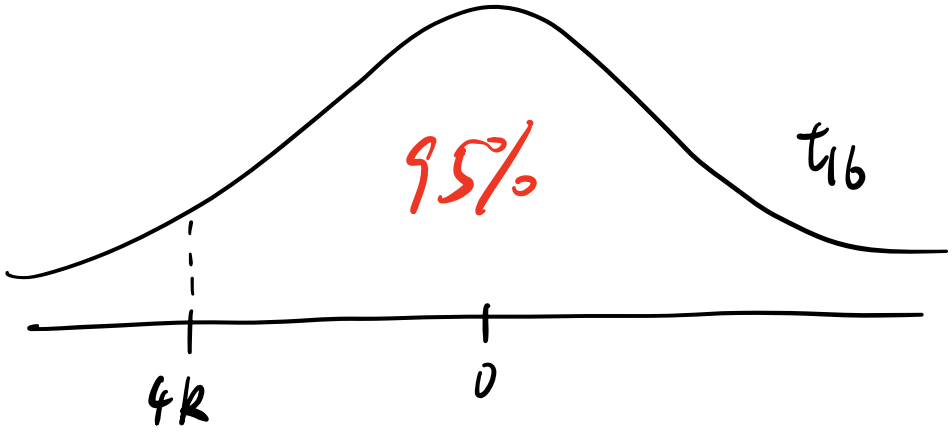
Let $T = \frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1} = t_{16}$

$$\mathbb{P}\left[\hat{\mu} > \mu + k\hat{\sigma}\right] = \mathbb{P}\left[\frac{\bar{X} - \mu}{\hat{\sigma}} > k\right]$$

$$= \mathbb{P}\left[\frac{\bar{X} - \mu}{\sqrt{\frac{n-1}{n}}\, S} > k\right] = \mathbb{P}\left[T > \sqrt{n-1}\, k\right]$$

Example

Want $P[T > \sqrt{16}\, k] = 0.95$



$$4k = t_{16}^{0.95} = -1.7459 \quad \text{(Computer)}$$

$$k \simeq -0.436$$